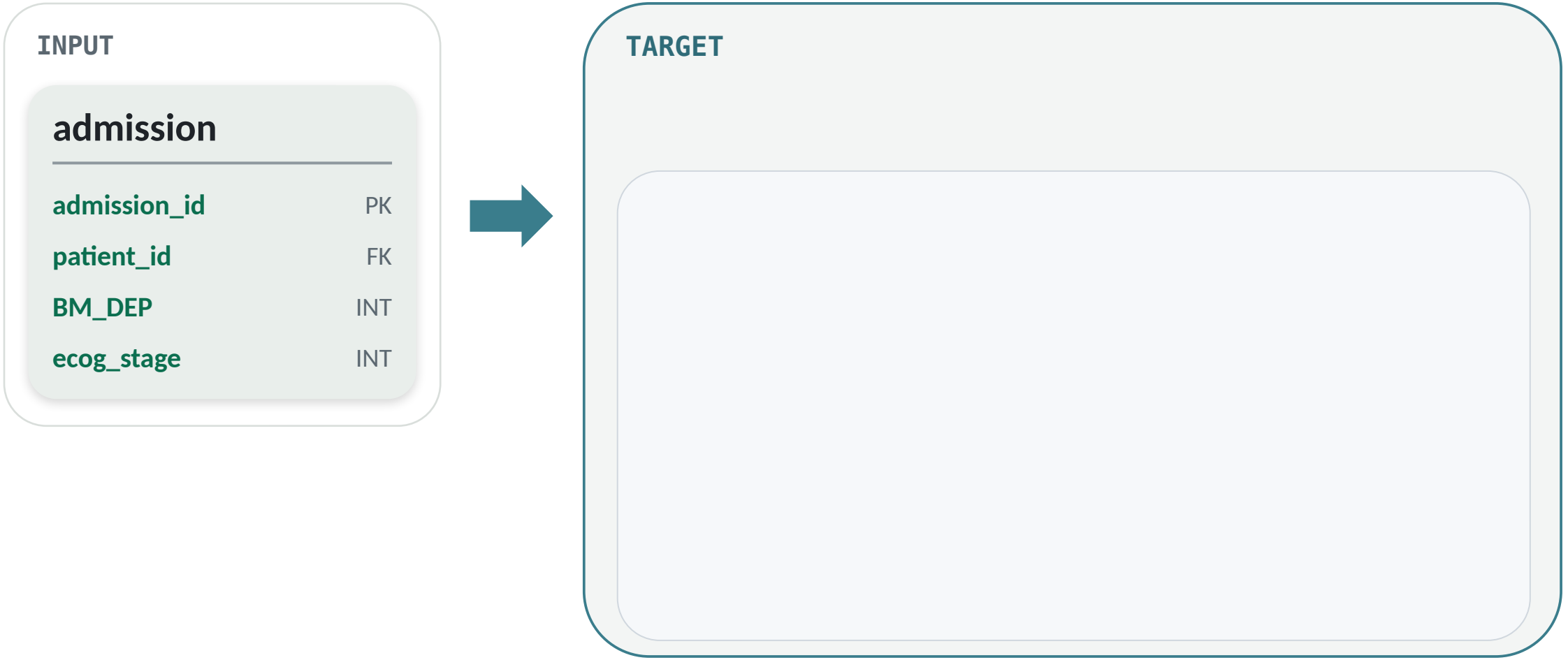
WHAT WE WANT TO BUILD

Not just a cleaner schema.
A usable semantic model.



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INPUT

admission

admission_id	PK
patient_id	FK
BM_DEP	INT
ecog_stage	INT



TARGET

Admission —hasPatientData→ PatientData

```
Class: Admission
  SubClassOf:
    admission_id some xsd:integer,
    hasPatientData some PatientData,        ← schema (foreign key)
    BM_DEP only xsd:integer,
    ecog_stage only xsd:integer

DataProperty: admission_id
ObjectProperty: hasPatientData              ← schema (foreign key)
DataProperty: BM_DEP
  Annotations:

    prov:wasDerivedFrom admission:BM_DEP    ← source column
DataProperty: ecog_stage
  Annotations:

    prov:wasDerivedFrom admission:ecog_stage ← source column
```

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  SubClassOf:
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    BM_DEP only xsd:integer,
    ecog_stage only xsd:integer

DataProperty: admission_id
ObjectProperty: hasPatientData                ← schema (foreign key)
DataProperty: BM_DEP
  Annotations:
    rdfs:label "Bone Marrow Depression"@en,   ← documentation

    prov:wasDerivedFrom admission:BM_DEP      ← source column
DataProperty: ecog_stage
  Annotations:
    rdfs:label "ECOG Performance Status"@en,  ← documentation

    prov:wasDerivedFrom admission:ecog_stage  ← source column
```

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  SubClassOf:
    admission_id some xsd:integer,
    hasPatientData some PatientData,          ← schema (foreign key)
    BM_DEP only xsd:integer,
    ecog_stage only xsd:integer

DataProperty: admission_id
ObjectProperty: hasPatientData                ← schema (foreign key)
DataProperty: BM_DEP
  Annotations:
    rdfs:label "Bone Marrow Depression"@en,   ← documentation
    skos:relatedMatch obo:DOID_4961,          ← Disease Ontology
    prov:wasDerivedFrom admission:BM_DEP      ← source column
DataProperty: ecog_stage
  Annotations:
    rdfs:label "ECOG Performance Status"@en,  ← documentation

    prov:wasDerivedFrom admission:ecog_stage  ← source column
```

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  SubClassOf:
    admission_id some xsd:integer,
    hasPatientData some PatientData,          ← schema (foreign key)
    BM_DEP only xsd:integer,
    ecog_stage only xsd:integer
  DisjointWith: PatientData                   ← LLM knowledge
DataProperty: admission_id
ObjectProperty: hasPatientData                ← schema (foreign key)
DataProperty: BM_DEP
  Annotations:
    rdfs:label "Bone Marrow Depression"@en,   ← documentation
    skos:relatedMatch obo:DOID_4961,          ← Disease Ontology
    prov:wasDerivedFrom admission:BM_DEP      ← source column
DataProperty: ecog_stage
  Annotations:
    rdfs:label "ECOG Performance Status"@en,  ← documentation
    rdfs:comment "0 = fully active … 5 = dead"@en,  ← LLM knowledge
    prov:wasDerivedFrom admission:ecog_stage  ← source column
```

WHAT WE WANT TO BUILD

Not just a cleaner schema. A usable semantic model.

INPUT

admission

admission_id	PK
patient_id	FK
BM_DEP	INT
ecog_stage	INT

WHAT THIS ENABLES

Ontology-based data access (OBDA)
Knowledge graph population
Semantic data integration

TARGET

Admission —hasPatientData→ PatientData

```
Class: Admission
  SubClassOf:
    admission_id some xsd:integer,
    hasPatientData some PatientData,        ← schema (foreign key)
    BM_DEP only xsd:integer,
    ecog_stage only xsd:integer
  DisjointWith: PatientData                 ← LLM knowledge
DataProperty: admission_id
ObjectProperty: hasPatientData              ← schema (foreign key)
DataProperty: BM_DEP
  Annotations:
    rdfs:label "Bone Marrow Depression"@en,     ← documentation
    skos:relatedMatch obo:DOID_4961,            ← Disease Ontology
    prov:wasDerivedFrom admission:BM_DEP        ← source column
DataProperty: ecog_stage
  Annotations:
    rdfs:label "ECOG Performance Status"@en,    ← documentation
    rdfs:comment "0 = fully active … 5 = dead"@en,  ← LLM knowledge
    prov:wasDerivedFrom admission:ecog_stage    ← source column
```